

Mohamad Awad

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Second-year civil engineering student at Toronto Metropolitan University, building automated infrastructure-inspection tools (AI + drone-aerial imagery) and bringing software-engineering depth to civil workflows.

Education

Bachelor of Engineering (BEng), Civil Engineering Sept 2024 – Jun 2028 Toronto Metropolitan University, Toronto, ON

Relevant Coursework: Project Management, Engineering Graphics, Engineering Design, Construction Materials

Technical Projects

PaveScan AI — Automated Pavement-Distress Inspection System

Independent Project, 2026 — [GitHub](#)

- Built a pavement-distress inspection system that takes road-surface images, runs AI crack detection, scores each section against the ASTM D6433 Pavement Condition Index (PCI) standard, and produces a maintenance-priority report — addressing a problem civil engineering firms currently solve with manual surveys at thousands of dollars per mile.
- Trained a YOLO11 segmentation model with SAHI tiled inference and Test-Time Augmentation at 1280 px input; fine-tuning achieved Box mAP50 of 0.816 and Mask mAP50 of 0.395 on a custom pavement dataset.
- Wrote an ASTM D6433 PCI scoring module that classifies severity by defect type, area, width, and detection confidence, and produces safety-prioritized inspection recommendations.
- Built a 4-page Streamlit dashboard with an interactive Folium GPS map showing severity heatmaps, plus a WeasyPrint PDF report generator with PCI scores and maintenance recommendations.
- Wrote the data and training pipeline: dataset prep, augmentation, YOLO11 fine-tuning with focal loss, model calibration, and an ensemble IoU deduplication step that combines two detectors.
- Diagnosed and fixed a numerical-stability failure during training where mixed-precision overflow was poisoning the EMA buffers and silently freezing checkpoint weights — root-caused from a single warning, fixed by disabling AMP and re-running from a clean baseline.
- Fine-tuned the segmentation model on cloud GPU infrastructure (Google Colab Pro A100), managing compute budget across multi-hour training runs.

Engineering Design Project — Mobile Storage & Mobility Aid

Toronto Metropolitan University, Feb – Mar 2025

- Designed a mobile storage cart prototype for unhoused individuals, optimizing for durability, weather resistance, and one-handed mobility.
- Co-designed a walking-cane prototype with a small team, evaluating user needs and structural stability against load and ergonomic constraints.

Pop-Up Chapel Co. — Platform Foundations Consulting Engagement

Email Automation & Document Generation Lead, Riipen Consulting Engagement, Apr – May 2026 (60 hrs)

- Built a Python pipeline (Pydantic + Jinja2 + Playwright) that auto-generates 8 branded wedding-day documents from booking data in roughly 10 seconds — replacing 8+ hours of manual document rebuilding per chapel event.
- Deployed a live Next.js + TypeScript companion tool on Vercel (popup-chapel-docs.vercel.app) with Google Sheets API integration, HMAC-signed cookie auth, Zod schema validation, and rate limiting.
- Mapped 49 client touchpoints across 9 lifecycle stages and identified 9 missing touchpoints, flagged by the CEO as the engagement's standout deliverable.
- Authored 5 platform specifications (communication audit, touchpoint map, email-template framework, automation opportunity map, communication strategy) — feeding a 12-state booking machine and Stripe webhook architecture for the team's upcoming production SaaS.
- Used the Anthropic Claude SDK (with prompt caching) for AI-assisted copy generation in the Python prototype, with a deterministic fallback path for offline runs.

Work Experience

Fitness Centre Desk Supervisor *May 2025 – Present* Toronto Metropolitan University, Toronto, ON

- Supervise daily facility operations for 35+ staff, tracking task completion against safety and operational procedures.
- Inspect 30+ pieces of equipment per shift and submit service requests for any maintenance issues.
- Resolve shift-level operational issues escalated by staff and coordinate with management on facility decisions.

Project Coordinator (Marketing) *Jun 2024 – Aug 2024* BrandEQ, Toronto, ON

- Coordinated project timelines for 6+ client accounts, tracking deliverables and providing weekly status updates.
- Wrote weekly performance reports for clients, summarizing campaign results and recommended next steps.

Technical Skills

- **Engineering software:** AutoCAD, Civil 3D, Autodesk Revit, SolidWorks, Microsoft Project
- **Programming & ML:** Python (Pydantic, Jinja2, Streamlit, Playwright, OpenCV, PyTorch, Ultralytics YOLO11, SAHI), TypeScript, Next.js (App Router), React, Tailwind CSS, shadcn/ui
- **Data & infrastructure:** PostgreSQL (data modeling), Google Sheets API, Stripe webhooks, Vercel, GitHub, Git
- **AI / automation tooling:** Anthropic Claude SDK (with prompt caching), Gemini API, Zod schema validation, HMAC-signed auth, agentic workflows, cloud GPU training (Google Colab Pro)
- **Office & documentation:** Microsoft Excel (advanced — data organization, spreadsheets, reporting), Word, PowerPoint